# Technology Stack

| | |
|---|---|
| **Date** | 03 July 2026 |
| **Team ID** | SL-2026-02 |
| **Project Name** | Smart Lender – Loan Eligibility Prediction Using Machine Learning |
| **Maximum Marks** | 2 Marks |

**Define Your Technology Stack:**

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

**Technology Stack Details:**

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| 1 | Frontend / Client-Side | HTML5, CSS3, Bootstrap 5, JavaScript | Provides a responsive and user-friendly interface for applicants to enter loan details and instantly view loan eligibility results across desktop and mobile devices. |
| 2 | Backend / Server-Side | Python, Flask Framework | Flask offers a lightweight and efficient web framework for integrating the trained machine learning model and processing prediction requests with minimal overhead. |

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| 3 | Database / Data Storage | CSV Dataset, Pandas DataFrames, Joblib (.pkl model files) | The project uses structured CSV data for training. Pandas efficiently handles preprocessing, while Joblib stores the trained ML model for fast loading during prediction. |
| 4 | Cloud / Hosting / Deployment | Local Flask Server (Development), Render/Heroku (Optional Deployment) | Flask enables quick local deployment for testing and demonstration. Cloud platforms such as Render or Heroku can be used for online deployment with minimal configuration. |
| 5 | Version Control & CI/CD | Git, GitHub | Git enables version tracking and collaboration among team members, while GitHub provides secure source code management and project documentation. |
| 6 | Third-Party APIs / Other Tools | Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, VS Code | Scikit-learn is used to train and evaluate machine learning models. Pandas and NumPy support data preprocessing, Matplotlib and Seaborn provide data visualization, Jupyter Notebook is used for experimentation, and VS Code is used for application development. |